NEET UG BIOLOGY • HUMAN PHYSIOLOGY

Body Fluids and Circulation

Comprehensive Study Tutorial, Conceptual Frameworks, Worked Examples, and NEET UG Practice Assessment

1. Learning Objectives and Overview

Every living cell requires a continuous supply of nutrients, O_2, and other essential substances, while metabolic wastes and harmful substances such as CO_2 must be removed continuously for healthy tissue functioning. Different animal groups have evolved distinct transport mechanisms:

- **Simple organisms (Sponges and Coelenterates):** Circulate water from their surroundings directly through their body cavities to facilitate cellular exchange.
- **Complex organisms (including Humans):** Utilize specialized internal body fluids—primarily **blood** and **lymph (tissue fluid)**—to transport materials throughout the body.

NEET UG Focus Areas in this Tutorial:

- Composition of blood: Plasma proteins, minerals, serum, and quantitative profiles of formed elements (RBCs, WBCs, and platelets).
- Immunological and clinical basis of ABO and Rh blood grouping, including *erythroblastosis foetalis*.
- Enzyme cascade of blood coagulation and the formation of lymph.
- Evolution of vertebrate hearts, open vs. closed circulation, and human cardiac anatomy & nodal conduction pathway.
- Phases of the cardiac cycle, heart sounds, cardiac output calculations, and standard Electrocardiogram (ECG) wave interpretation.
- Double circulation, portal systems, neuro-hormonal regulation of cardiac activity, and circulatory disorders.

2. Core Concepts and Detailed Explanations

2.1 Blood: A Special Connective Tissue

Blood is a specialized fluid connective tissue consisting of a fluid matrix called **plasma** ($\approx 55\%$) and cellular components called **formed elements** ($\approx 45\%$). It is considered a connective tissue because it originates from the mesoderm, consists of cells suspended in an abundant extracellular matrix (plasma), and connects various organ systems across the body by transporting gases, nutrients, hormones, and wastes.

A. Plasma (55% of Blood)

Plasma is a straw-coloured, viscous fluid. Its composition includes:

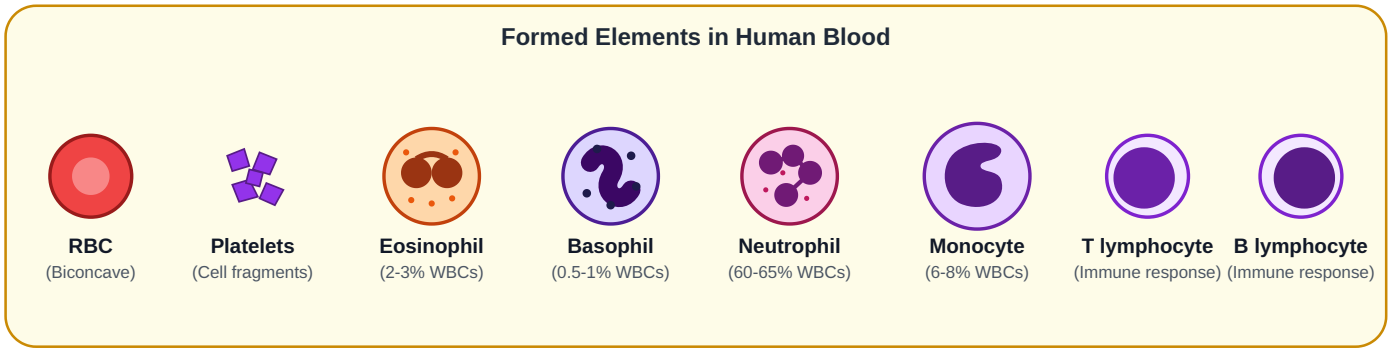
- **Water:** Constitutes $90–92\%$ of plasma.
- **Proteins:** Contribute $6–8\%$ of plasma. The three major plasma proteins are:
  - **Fibrinogens:** Essential for the clotting or coagulation of blood.
  - **Globulins:** Primarily involved in the defense mechanisms of the body (acting as antibodies).
  - **Albumins:** Maintain the osmotic balance of blood.
- **Minerals (Ions):** Present in small amounts, such as Na^+, Ca^{++}, Mg^{++}, HCO_3^-, and Cl^-.
- **Nutrients in Transit:** Glucose, amino acids, lipids, etc., are always present in plasma as they travel between tissues.
- **Clotting Factors:** Present in plasma in an *inactive* form.

Key Definition — Serum: Plasma without the clotting factors is called **serum**.

$$\text{Serum} = \text{Plasma} - \text{Clotting Factors (including Fibrinogen)}$$

B. Formed Elements (45% of Blood)

Erythrocytes, leucocytes, and platelets are collectively called formed elements.



Tutorial Diagram 1: Diagrammatic representation of the formed elements in blood.

Formed Element	Normal Count (per mm^3 of blood)	Structural Features & Site of Formation	Major Functions & Lifespan
Erythrocytes (RBCs)	5 to 5.5 million mm^{-3} (Most abundant blood cells)	Biconcave shape; devoid of nucleus in most mammals; formed in red bone marrow in adults. Contains red iron-containing complex protein **haemoglobin** ($12–16$ g per 100 mL of blood).	Significant role in transport of respiratory gases (O_2 and CO_2). Average lifespan is **120 days**, after which they are destroyed in the **spleen** (graveyard of RBCs).
Leucocytes (WBCs)	$6000–8000\ mm^{-3}$	Colourless (lack haemoglobin); **nucleated**; divided into Granulocytes (neutrophils, eosinophils, basophils) and Agranulocytes (lymphocytes, monocytes).	Generally short-lived; primary defense and immune cells of the body.
Platelets (Thrombocytes)	$1,50,000–3,50,000\ mm^{-3}$	Cell fragments produced from **megakaryocytes** (special cells in the bone marrow).	Release a variety of substances involved in coagulation or clotting of blood. Reduction in number causes clotting disorders and excessive blood loss.

Detailed Classification of Leucocytes (WBCs)

- **Granulocytes:**
  - **Neutrophils ($60–65\%$ of total WBCs):** Most abundant WBCs. They are **phagocytic cells** that destroy foreign organisms entering the body.
  - **Eosinophils ($2–3\%$ of total WBCs):** Resist infections and are associated with **allergic reactions**.
  - **Basophils ($0.5–1\%$ of total WBCs):** Least abundant WBCs. They secrete **histamine, serotonin, heparin**, etc., and are involved in **inflammatory reactions**.
- **Agranulocytes:**
  - **Lymphocytes ($20–25\%$ of total WBCs):** Exist in two major types—**'B'** and **'T'** forms. Both B and T lymphocytes are responsible for the **immune responses** of the body.
  - **Monocytes ($6–8\%$ of total WBCs):** Along with neutrophils, they are **phagocytic cells** that destroy foreign pathogens.

2.2 Blood Groups: ABO and Rh Systems

A. ABO Grouping

ABO grouping is based on the presence or absence of two surface antigens (chemicals that can induce an immune response) on the RBCs, namely **Antigen A** and **Antigen B**. Correspondingly, the plasma of different individuals contains two natural antibodies (proteins produced in response to antigens): **anti-A** and **anti-B**.

Before any blood transfusion, donor blood must be carefully matched with recipient blood to prevent severe **clumping (agglutination and destruction of RBCs)**.

Blood Group	Antigens on RBCs	Antibodies in Plasma	Compatible Donor's Group	Special Status
A	A	anti-B	A, O	—
B	B	anti-A	B, O	—
AB	A, B	nil	AB, A, B, O	**Universal Recipient** (lacks plasma antibodies against A and B)
O	nil	anti-A, B	O	**Universal Donor** (lacks A and B surface antigens on RBCs)

B. Rh Grouping and Erythroblastosis Foetalis

The **Rh antigen** (named after a similar antigen first discovered in Rhesus monkeys) is present on the surface of RBCs in the majority (nearly 80%) of humans. Individuals possessing this antigen are **Rh positive ($\mathrm{Rh}+\mathrm{ve}$)**, while those lacking it are **Rh negative ($\mathrm{Rh}-\mathrm{ve}$)**. If an $\mathrm{Rh}-\mathrm{ve}$ person is exposed to $\mathrm{Rh}+\mathrm{ve}$ blood, they form specific antibodies against the Rh antigens.

Clinical Case — Erythroblastosis Foetalis (Rh Incompatibility):

- **Condition:** Occurs when a pregnant mother is $\mathrm{Rh}-\mathrm{ve}$ and the foetus is $\mathrm{Rh}+\mathrm{ve}$.
- **First Pregnancy:** Foetal Rh antigens do not get exposed to the mother's $\mathrm{Rh}-\mathrm{ve}$ blood because the two bloodstreams are well separated by the placenta. Thus, the first child is born safely.
- **Sensitization at Delivery:** During delivery of the first $\mathrm{Rh}+\mathrm{ve}$ child, maternal blood may be exposed to small amounts of $\mathrm{Rh}+\mathrm{ve}$ foetal blood. The mother then begins producing anti-Rh antibodies in her blood.
- **Subsequent Pregnancies:** If a subsequent foetus is also $\mathrm{Rh}+\mathrm{ve}$, the mother's Rh antibodies can leak across the placenta into the foetal blood and destroy the foetal RBCs. This can be fatal to the foetus or cause **severe anaemia and jaundice** in the baby, a condition termed *erythroblastosis foetalis*.
- **Prophylaxis (Prevention):** Administering **anti-Rh antibodies** to the mother immediately after the delivery of the first child destroys any leaked foetal $\mathrm{Rh}+\mathrm{ve}$ RBCs before the mother's immune system gets sensitized.

2.3 Coagulation of Blood

In response to an injury or trauma, blood exhibits coagulation or clotting to prevent excessive blood loss. The dark reddish-brown scum formed over a wound is a **clot or coagulum** composed mainly of a network of insoluble threads called **fibrins** in which dead and damaged formed elements of blood are trapped.

1. **Initiation:** An injury or trauma stimulates blood platelets to release certain coagulation-activating factors. Damaged tissues at the site of injury also release factors that initiate coagulation.
2. **Cascade Process:** Inactive clotting factors in plasma undergo a series of linked enzymic reactions (a **cascade process**) in the presence of **calcium ions (Ca^{++})** to form an enzyme complex called **thrombokinase**.
3. **Prothrombin Activation:** Thrombokinase (along with Ca^{++}) converts the inactive plasma protein **prothrombin** into the active enzyme **thrombin**.
4. **Fibrin Formation:** Thrombin catalyses the conversion of soluble, inactive **fibrinogens** in the plasma into insoluble **fibrins**, which polymerize into a meshwork and trap blood cells to form the clot.

$$\text{Injured Tissue} + \text{Platelets} \xrightarrow{\text{Cascade Process } (\mathrm{Ca}^{++})} \text{Thrombokinase}$$

$$\text{Prothrombin (Inactive)} \xrightarrow{\text{Thrombokinase, } \mathrm{Ca}^{++}} \text{Thrombin (Active Enzyme)}$$

$$\text{Fibrinogen (Soluble)} \xrightarrow{\text{Thrombin}} \text{Fibrin Threads} + \text{Trapped Blood Cells} \longrightarrow \text{Clot (Coagulum)}$$

2.4 Lymph (Tissue Fluid)

As blood flows through tissue capillaries, some water along with many small water-soluble substances filter out into the intercellular spaces of tissues, leaving the **larger proteins** and **most of the formed elements** inside the blood vessels. This released fluid is called **interstitial fluid** or **tissue fluid**.

- **Composition:** It has the **same mineral distribution** as plasma, but lacks RBCs, platelets, and large plasma proteins.
- **Function of Tissue Fluid:** All exchange of nutrients, gases (O_2, CO_2), etc., between the blood and tissue cells always occurs through this interstitial fluid.
- **Lymphatic System:** An elaborate network of vessels collects this interstitial fluid and drains it back into the **major veins**. Once inside the lymphatic system, the fluid is called **lymph**.
- **Properties & Roles of Lymph:**
  - Colourless fluid containing specialized **lymphocytes** responsible for immune responses.
  - Important carrier for nutrients, hormones, etc.
  - **Fat Absorption:** Fats are absorbed through lymph in the **lacteals** present in the intestinal villi.

2.5 Circulatory Pathways Across the Animal Kingdom

Feature	Open Circulatory System	Closed Circulatory System
Mechanism	Blood pumped by the heart passes through large vessels into open spaces or body cavities called **sinuses**.	Blood pumped by the heart is always circulated through a closed network of blood vessels (arteries, capillaries, veins).
Regulation	Fluid flow is slower and less precisely regulated.	More advantageous because the flow of fluid can be much more precisely regulated.
Examples	**Arthropods** and **Molluscs**.	**Annelids** and **Chordates**.

Evolutionary Pattern of Vertebrate Hearts and Circulation

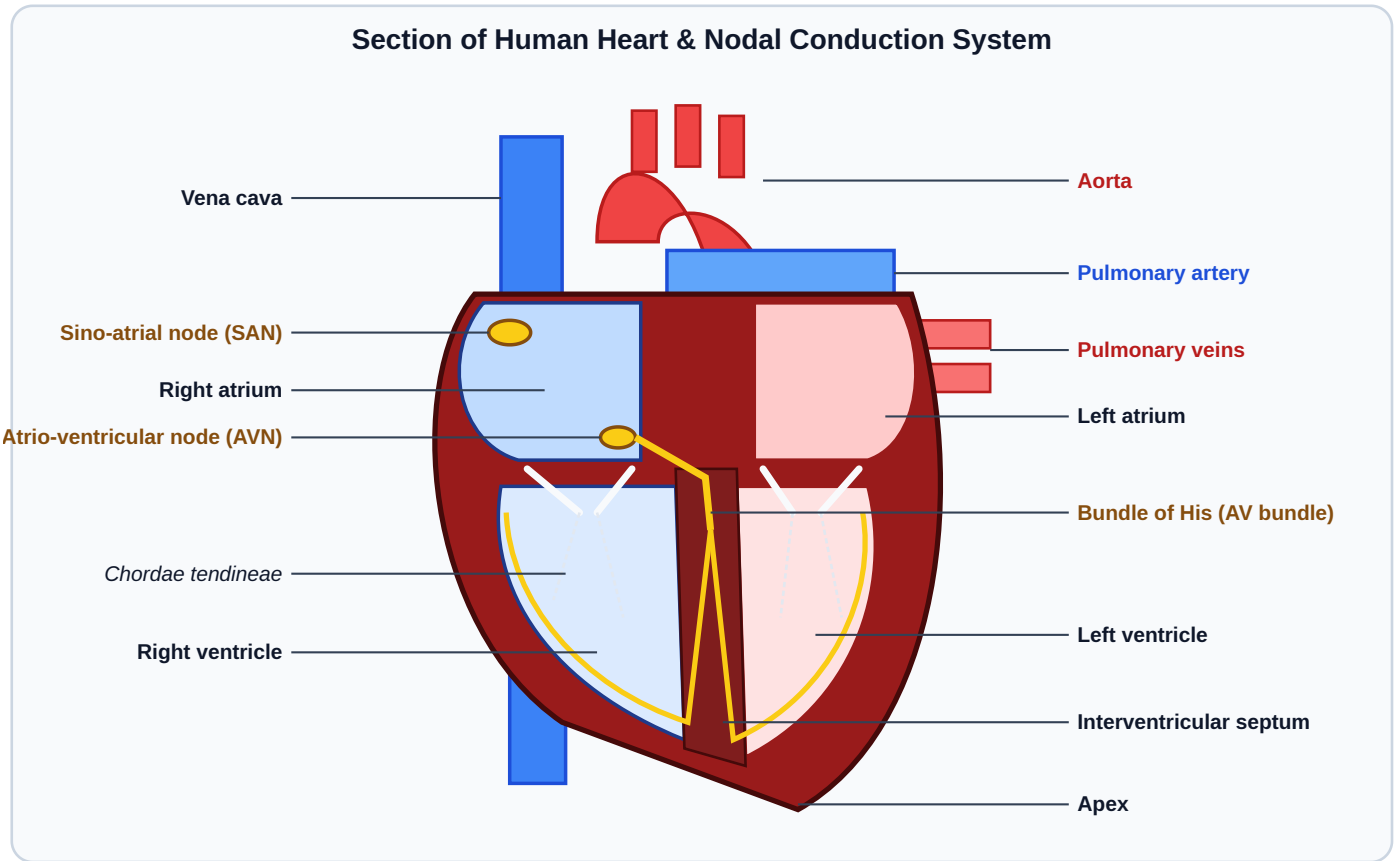
All vertebrates possess a muscular chambered heart:

- **Fishes (2-Chambered Heart):** Possess 1 atrium and 1 ventricle. The heart pumps out deoxygenated blood, which is oxygenated by the gills and supplied directly to the body parts, from where deoxygenated blood returns to the heart. This pathway is called **single circulation**.
- **Amphibians and Reptiles, except Crocodiles (3-Chambered Heart):** Possess 2 atria and 1 single ventricle. The left atrium receives oxygenated blood from the gills/lungs/skin, and the right atrium receives deoxygenated blood from other body parts. However, the two streams get mixed up in the single ventricle, which pumps out mixed blood. This is called **incomplete double circulation**.
- **Crocodiles, Birds, and Mammals (4-Chambered Heart):** Possess 2 atria and 2 ventricles. Oxygenated blood (received by the left atrium) and deoxygenated blood (received by the right atrium) pass into the ventricles of the same sides and are pumped out without any mixing. Because two completely separate circulatory pathways exist, these animals exhibit complete **double circulation**.

2.6 Human Circulatory System & Heart Anatomy

The human blood vascular system consists of a muscular chambered heart, a closed branching network of blood vessels, and blood.

- **Origin & Location:** The heart is a **mesodermally derived** organ situated in the thoracic cavity, in between the two lungs, slightly tilted to the left. It is the size of a clenched fist.
- **Protection:** Enclosed by a double-walled membranous bag called the **pericardium**, which contains the **pericardial fluid**.
- **Chambers & Septa:** Four chambers—two relatively small upper **atria** (thin-walled) and two larger, much thicker-walled lower **ventricles**.
  - **Inter-atrial septum:** A thin muscular wall separating the right and left atria.
  - **Inter-ventricular septum:** A thick-walled muscular partition separating the left and right ventricles.
  - **Atrio-ventricular septum:** Thick fibrous tissue separating the atrium and ventricle of the same side; each possesses an opening connecting the two ipsilateral chambers.
- **Heart Valves (Prevent Backward Flow & Ensure Unidirectional Flow):**
  - **Tricuspid valve:** Formed of three muscular flaps or cusps; guards the opening between the **right atrium and right ventricle**.
  - **Bicuspid (or Mitral) valve:** Guards the opening between the **left atrium and left ventricle**.
  - **Semilunar valves:** Guard the openings of the **right ventricle into the pulmonary artery** and the **left ventricle into the aorta**.



Tutorial Diagram 2: Internal section of the human heart showing chambers, valves, major vessels, and the autoexcitable nodal conduction tissue.

Nodal Tissue and the Conduction Pathway

A specialized cardiac musculature called the **nodal tissue** is distributed in the heart. It has the intrinsic ability to generate action potentials without any external stimuli, meaning it is **autoexcitable**:

1. **Sino-atrial node (SAN):** Located in the **right upper corner of the right atrium**. It generates the maximum number of action potentials ($70–75\ \text{min}^{-1}$) and initiates and maintains the rhythmic contractile activity of the heart. Hence, the SAN is called the **pacemaker**.

2. **Atrio-ventricular node (AVN):** Located in the **lower left corner of the right atrium** close to the atrio-ventricular septum.
3. **Atrio-ventricular bundle (AV bundle / Bundle of His):** Continues from the AVN, passes through the atrio-ventricular septa to emerge on the top of the inter-ventricular septum, and immediately divides into a **right and left bundle**.
4. **Purkinje Fibres:** Minute fibres arising from the right and left bundle branches that spread throughout the ventricular musculature of their respective sides.

2.7 Cardiac Cycle, Heart Sounds, and Electrocardiogram (ECG)

A. Phases of the Cardiac Cycle

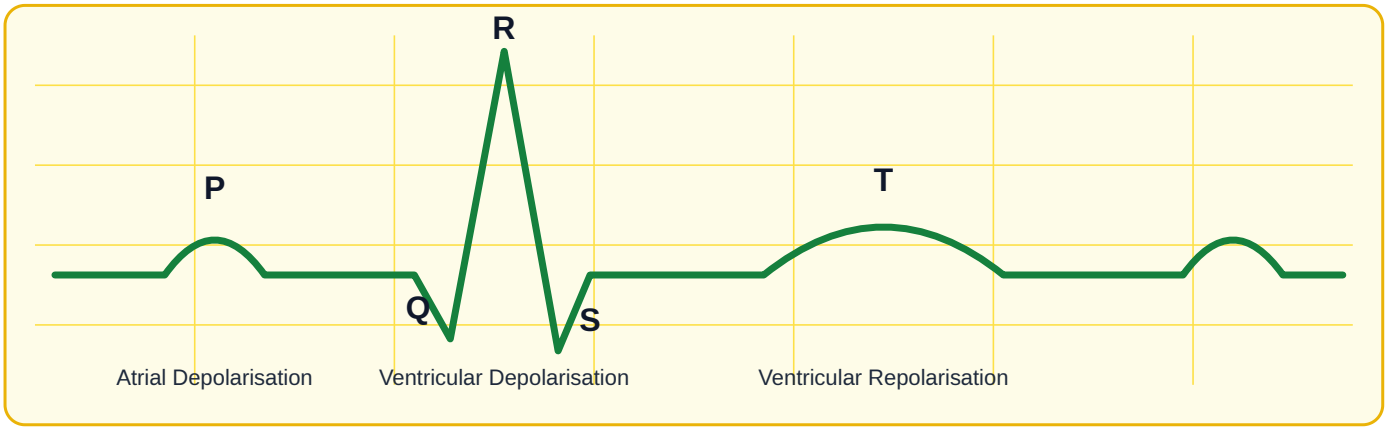
The sequential event in the heart that is cyclically repeated is called the **cardiac cycle**, consisting of systole (contraction) and diastole (relaxation) of both atria and ventricles. At an average heart rate of $72\ \text{beats}\ \text{min}^{-1}$, the duration of one cardiac cycle is $0.8\ \text{seconds}$.

Stage of Cardiac Cycle	Chamber State & Pressure Changes	Position of Valves	Blood Flow & Key Events
1. Joint Diastole	All four chambers (both atria and both ventricles) are in a relaxed state.	**Tricuspid & Bicuspid valves:** Open **Semilunar valves:** Closed	Blood from pulmonary veins and vena cava flows passively into the left and right ventricles respectively through the left and right atria ($\approx 70\%$ of ventricular filling).
2. Atrial Systole	SAN generates an action potential stimulating both atria to undergo simultaneous contraction.	**Tricuspid & Bicuspid valves:** Open **Semilunar valves:** Closed	Increases the flow of blood into the ventricles by about 30%.
3. Ventricular Systole (Atrial Diastole)	Action potential reaches ventricles via AVN, AV bundle (Bundle of His), and Purkinje fibres. Ventricles contract while atria relax simultaneously. Ventricular pressure rises sharply.	**Tricuspid & Bicuspid valves:** Forced **Closed** (prevents backflow into atria; produces **1st heart sound "lub"**). **Semilunar valves:** Forced **Open** as ventricular pressure rises further.	Blood is pumped out from the right ventricle into the pulmonary artery and from the left ventricle into the aorta.
4. Ventricular Diastole	Ventricles relax and ventricular pressure falls rapidly, then declines further below atrial pressure.	**Semilunar valves:** Close initially (prevents backflow into ventricles; produces **2nd heart sound "dub"**). **Tricuspid & Bicuspid valves:** Pushed **Open** by atrial blood pressure as ventricular pressure declines further.	Blood emptied into the atria by the veins once again moves freely into the ventricles, returning the heart to **joint diastole**.

B. Electrocardiogram (ECG)

An **electrocardiogram (ECG)** is a graphical representation of the electrical activity of the heart during a cardiac cycle, recorded by an instrument called the **electrocardiograph**.

- **Standard Lead Placement:** A patient is connected to the machine with **three electrical leads**—one to **each wrist** and one to the **left ankle**—to continuously monitor heart activity. For a detailed evaluation of heart function, multiple leads are attached to the chest region.
- **Clinical Utility:** Counting the number of **QRS complexes** in a given time period determines the **heart beat rate** of an individual. Any deviation from the standard waveform indicates a possible abnormality or disease.



Tutorial Diagram 3: Diagrammatic presentation of a standard Electrocardiogram (ECG).

- **P-wave:** Represents the **electrical excitation (or depolarisation) of the atria**, which leads to the contraction of both the atria (atrial systole).
- **QRS complex:** Represents the **depolarisation of the ventricles**, which initiates ventricular contraction. The contraction starts shortly after Q and marks the **beginning of the systole**.
- **T-wave:** Represents the return of the ventricles from the excited to the normal state (**repolarisation**). The **end of the T-wave marks the end of systole**.

2.8 Double Circulation and Vascular Histology

A. Three Layers of Blood Vessels (Arteries and Veins)

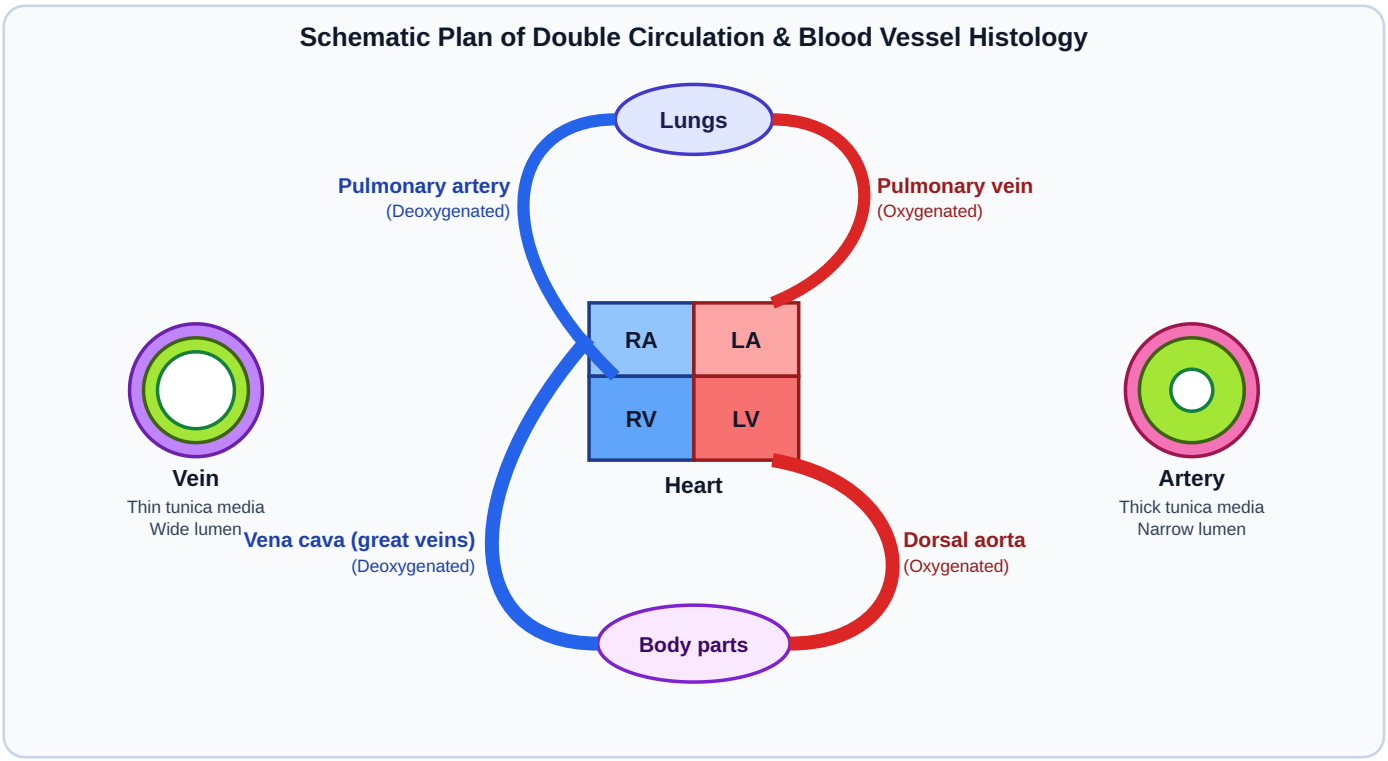
1. **Tunica intima (Inner layer):** Inner lining of **squamous endothelium**.
2. **Tunica media (Middle layer):** Composed of **smooth muscle and elastic fibres**. Crucially, the *tunica media is comparatively thin in veins* (resulting in a wider lumen in veins compared to arteries).
3. **Tunica externa (External layer):** Composed of **fibrous connective tissue with collagen fibres**.

B. Pulmonary, Systemic, Hepatic Portal, and Coronary Circulations

- **Pulmonary Circulation:**
  $$\text{Right Ventricle (Deoxygenated)} \longrightarrow \text{Pulmonary Artery} \longrightarrow \text{Lungs (Oxygenation)} \longrightarrow \text{Pulmonary Veins (Oxygenated)} \longrightarrow \text{Left Atrium}$$
- **Systemic Circulation:**
  $$\text{Left Ventricle (Oxygenated)} \longrightarrow \text{Aorta} \longrightarrow \text{Arteries, Arterioles, Capillaries} \longrightarrow \text{Tissues (Exchange)} \longrightarrow \text{Venules, Veins, Vena Cava (Deoxy}$$

Significance: Provides nutrients, O_2, and essential substances to tissues while taking CO_2 and harmful substances away for elimination, without any mixing of oxygenated and deoxygenated blood.

- **Hepatic Portal System:** A unique vascular connection between the **digestive tract and the liver**. The **hepatic portal vein** carries nutrient-rich blood from the intestine to the liver before it is delivered to the systemic circulation.
- **Coronary System:** A special system of blood vessels present exclusively for the circulation of blood to and from the **cardiac musculature** (heart muscle).



Tutorial Diagram 4: Schematic plan of human double circulation (pulmonary and systemic pathways) and comparative cross-sections of a vein and an artery.

2.9 Regulation of Cardiac Activity

- **Intrinsic (Myogenic) Regulation:** Normal activities of the human heart are regulated intrinsically (auto-regulated) by specialized muscles (**nodal tissue**). Therefore, the human heart is called **myogenic**.
- **Extrinsic Neural & Hormonal Moderation:** A special neural centre in the **medulla oblongata** can moderate cardiac function through the **Autonomic Nervous System (ANS)**:
  - **Sympathetic neural signals:** Increase the rate of heart beat, increase the strength of ventricular contraction, and thereby **increase the cardiac output**.
  - **Parasympathetic neural signals:** Decrease the rate of heart beat, decrease the speed of conduction of action potentials, and thereby **decrease the cardiac output**.
  - **Adrenal medullary hormones (Adrenaline / Noradrenaline):** Also **increase the cardiac output**.

2.10 Disorders of the Circulatory System

Disorder	Pathophysiology, Diagnostic Values & Clinical Distinctions
High Blood Pressure (Hypertension)	Normal blood pressure is $120/80\ \text{mm Hg}$, where $120\ \text{mm Hg}$ is the **systolic (pumping) pressure** and $80\ \text{mm Hg}$ is the **diastolic (resting) pressure**. If repeated checks of an individual's blood pressure are $140/90\ \text{mm Hg}$ **or higher**, it indicates **hypertension**. It leads to heart diseases and damages vital organs like the **brain and kidney**.
Coronary Artery Disease (CAD) / Atherosclerosis	Affects the coronary vessels that supply blood to the heart muscle. Caused by deposits of **calcium, fat, cholesterol, and fibrous tissues**, which makes the lumen of arteries narrower.
Angina (Angina Pectoris)	A symptom of **acute chest pain** appearing when **not enough oxygen is reaching the heart muscle**. Can occur in men and women of any age, but is more common among the middle-aged and elderly due to conditions affecting blood flow.
Heart Failure	The state of the heart when it is **not pumping blood effectively enough to meet the needs of the body**. Often called **congestive heart failure** because congestion of the lungs is one of its main symptoms. *Crucial Distinctions:* • **Heart Failure:** Ineffective pumping to meet body needs. • **Cardiac Arrest:** When the heart completely **stops beating**. • **Heart Attack:** When the heart muscle is **suddenly damaged** by an inadequate blood supply.

3. Key Equations and Quantitative Summary

1. Duration of One Cardiac Cycle

$$\text{Duration of Cardiac Cycle (seconds)} = \frac{60\ \text{seconds}}{\text{Heart Rate (beats per minute)}}$$

For a healthy adult with an average heart rate of $72\ \text{beats}\,\text{min}^{-1}$:

$$\text{Cardiac Cycle Duration} = \frac{60}{72} \approx 0.8\ \text{seconds}$$

2. Stroke Volume (Beat Volume)

The volume of blood pumped out by **each ventricle** during a single cardiac cycle: $\approx 70\,\text{mL}$.

3. Cardiac Output

The volume of blood pumped out by **each ventricle per minute**:

$$\text{Cardiac Output} = \text{Stroke Volume} \times \text{Heart Rate}$$
$$\text{Standard Cardiac Output} = 70\,\text{mL} \times 72\,\text{min}^{-1} = 5040\,\text{mL min}^{-1} \approx 5000\,\text{mL or } 5\,\text{litres min}^{-1}$$

4. Descending Order of WBC Abundance in Human Blood

$$\text{Neutrophils } (60\text{–}65\%) > \text{Lymphocytes } (20\text{–}25\%) > \text{Monocytes } (6\text{–}8\%) > \text{Eosinophils } (2\text{–}3\%) > \text{Basophils } (0.5\text{–}1\%)$$

4. Conceptual Connections and Comparative Analysis

Below are high-yield comparative summaries synthesizing core physiological distinctions frequently tested in competitive medical examinations:

Basis of Comparison	Blood	Lymph (Tissue Fluid)
Colour & RBCs	Red-coloured vascular fluid containing abundant erythrocytes (RBCs) and haemoglobin.	Colourless fluid completely devoid of RBCs and haemoglobin.
Protein & Cellular Content	Contains high concentration of large plasma proteins (6–8%) and all formed elements (RBCs, WBCs, platelets).	Lacks larger plasma proteins and most formed elements; contains specialized lymphocytes and the same mineral distribution as plasma.
Pathway	Circulates through arteries, capillaries, and veins pumped by the heart.	Forms in intercellular tissue spaces, enters lymphatic vessels, and drains back into major veins.

Basis of Comparison	Systole	Diastole
Mechanical State	Phase of active muscular **contraction** of cardiac chambers (atria or ventricles).	Phase of muscular **relaxation** and dilation of cardiac chambers.
Chamber Pressure & Blood Flow	Intra-chamber pressure increases; blood is actively ejected out of the chamber (from atria to ventricles, or from ventricles into the aorta and pulmonary artery).	Intra-chamber pressure falls; chambers receive and fill with blood arriving from veins or atria.
Arterial Blood Pressure	Corresponds to pumping pressure ($120\,\text{mm Hg}$ in a normal adult).	Corresponds to resting pressure ($80\,\text{mm Hg}$ in a normal adult).

5. Worked Examples

Worked Example 1: Athletic Cardiac Output and Cycle Duration

Problem: During intense physical exercise, an athlete's left ventricle pumps out $110\,\text{mL}$ of blood per cardiac cycle, and their ECG recording shows 30 QRS complexes in 15 seconds. Calculate (a) the athlete's heart rate per minute, (b) the duration of one cardiac cycle, and (c) the cardiac output in litres per minute.

Solution:

1. **Heart Rate:** Each QRS complex represents one ventricular depolarisation (one heartbeat).
$$\text{Heart Rate} = \frac{30\ \text{beats}}{15\ \text{seconds}} \times 60\ \text{seconds min}^{-1} = 120\ \text{beats min}^{-1}$$
2. **Duration of One Cardiac Cycle:**
$$\text{Duration} = \frac{60\ \text{seconds}}{120\ \text{beats min}^{-1}} = 0.5\ \text{seconds}$$
3. **Cardiac Output:** Given stroke volume $= 110\,\text{mL}$:
$$\text{Cardiac Output} = \text{Stroke Volume} \times \text{Heart Rate} = 110\,\text{mL} \times 120\,\text{min}^{-1} = 13,200\,\text{mL min}^{-1} = 13.2\ \text{litres min}^{-1}$$

Takeaway: As stated in physiology, the body can alter both stroke volume and heart rate so that the cardiac output of an athlete is much higher than that of an ordinary person.

Worked Example 2: ABO and Rh Blood Transfusion Compatibility

Problem: A trauma patient enters the emergency room and is typed as blood group $\text{B} - \text{ve}$ (B Rh-negative). Which surface antigens and plasma antibodies are naturally present in this patient's blood, and which blood groups from the set $\{\text{B} + \text{ve}, \text{B} - \text{ve}, \text{AB} - \text{ve}, \text{O} + \text{ve}, \text{O} - \text{ve}\}$ can be safely transfused to this patient?

Solution:

- **Antigens on patient's RBCs:** Only **Antigen B** is present (Rh antigen is absent because the patient is $\text{Rh} - \text{ve}$).
- **Natural antibodies in patient's plasma: anti-A** antibodies.
- **Compatibility Analysis:**
  - Because the patient has **anti-A** in plasma, donor RBCs must NOT have Antigen A. This rules out $\text{AB} - \text{ve}$. Both Group B and Group O lack Antigen A.
  - Because the patient is $\text{Rh} - \text{ve}$, exposure to $\text{Rh} + \text{ve}$ blood will induce the formation of specific anti-Rh antibodies. Therefore, only $\text{Rh} - \text{ve}$ donor blood should be transfused. This rules out $\text{B} + \text{ve}$ and $\text{O} + \text{ve}$.
- **Safe Donor Groups:** $\text{B} - \text{ve}$ and $\text{O} - \text{ve}$.

Worked Example 3: Active vs. Passive Ventricular Filling

Problem: In a healthy adult whose stroke volume is $70\,\text{mL}$ and end-diastolic ventricular volume increases by $70\,\text{mL}$ during each cycle, how much blood enters the ventricle during atrial systole compared to passive filling during joint diastole?

Solution:

- During joint diastole, blood from the pulmonary veins and vena cava flows passively through the open bicuspid and tricuspid valves into the ventricles, accounting for approximately 70% of ventricular filling:

$$\text{Passive filling volume} = 0.70 \times 70\,\text{mL} = 49\,\text{mL}$$

- When the SAN generates an action potential, simultaneous contraction of both atria (**atrial systole**) increases the flow of blood into the ventricles by about 30%:

$$\text{Active filling volume during atrial systole} = 0.30 \times 70\,\text{mL} = 21\,\text{mL}$$

6. NEET UG Practice Questions with Worked Solutions

Note: The following are original practice questions designed strictly in accordance with the NEET UG examination standard to test conceptual clarity, factual precision, and analytical application.

Practice Question 1 (Single-Choice MCQ)

Match List-I (Blood Component) with List-II (Primary Physiological Function) and select the correct option:

List-I (Blood Component)	List-II (Primary Physiological Function)
(A) Albumins	(i) Defense mechanisms of the body
(B) Globulins	(ii) Inflammatory reactions (secretes histamine, serotonin, heparin)
(C) Basophils	(iii) Osmotic balance in blood
(D) Eosinophils	(iv) Resist infections and associated with allergic reactions

(1) A-(i), B-(iii), C-(ii), D-(iv)

(2) A-(iii), B-(i), C-(ii), D-(iv)

(3) A-(iii), B-(i), C-(iv), D-(ii)

(4) A-(iv), B-(ii), C-(i), D-(iii)

Correct Answer: (2)

Worked Solution: Among the major plasma proteins (6–8% of plasma), **albumins** maintain osmotic balance (iii), **globulins** are primarily involved in defense mechanisms of the body (i), and fibrinogens are needed for clotting. Among granulocytes, **basophils** (0.5–1%) secrete histamine, serotonin, and heparin and are involved in inflammatory reactions (ii), while **eosinophils** (2–3%) resist infections and are associated with allergic reactions (iv).

Practice Question 2 (Multiple-Statement Evaluation)

Read the following statements regarding human formed elements carefully:

- (a) Erythrocytes in adult humans are nucleated, biconcave cells formed in the spleen.
- (b) A healthy individual has 12–$16\,\text{g}$ of haemoglobin in every $100\,\text{mL}$ of blood.
- (c) Neutrophils and monocytes are phagocytic cells that destroy foreign organisms entering the body.
- (d) Platelets are nucleated cells produced from megakaryocytes in the liver.
- (e) The correct ascending order of percentage of leucocytes in human blood is: Basophils < Eosinophils < Monocytes < Lymphocytes < Neutrophils.

Which of the above statements are **correct**?

(1) (a), (b), and (c) only

(2) (b), (c), and (e) only

(3) (b), (d), and (e) only

(4) (a), (c), (d), and (e) only

Correct Answer: (2)

Worked Solution:

- Statement (a) is incorrect: Mammalian RBCs are **devoid of a nucleus** and are formed in the **red bone marrow** in adults; they are *destroyed* in the spleen after an average lifespan of 120 days.
- Statement (b) is correct: Normal haemoglobin concentration is 12–$16\,\text{g}$ per $100\,\text{mL}$ of blood.
- Statement (c) is correct: Both neutrophils (60–65%) and monocytes (6–8%) are phagocytic cells.
- Statement (d) is incorrect: Platelets (thrombocytes) are non-nucleated **cell fragments** produced from megakaryocytes (special cells in the **bone marrow**, not liver).
- Statement (e) is correct: Basophils (0.5–1%) < Eosinophils (2–3%) < Monocytes (6–8%) < Lymphocytes (20–25%) < Neutrophils (60–65%).

Practice Question 3 (Assertion-Reason)

Assertion (A): The condition of *erythroblastosis foetalis* typically does not affect the first $\mathrm{Rh}+\mathrm{ve}$ child of an $\mathrm{Rh}-\mathrm{ve}$ mother, but can cause severe anaemia and jaundice in subsequent $\mathrm{Rh}+\mathrm{ve}$ pregnancies.

Reason (R): During the first pregnancy, the $\mathrm{Rh}+\mathrm{ve}$ blood of the foetus and the $\mathrm{Rh}-\mathrm{ve}$ blood of the mother are well separated by the placenta, and maternal sensitization usually occurs only during the delivery of the first child.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

Correct Answer: (1)

Worked Solution: Both Assertion and Reason are true, and Reason directly explains the Assertion. In the first pregnancy, the placenta separates the foetal $\mathrm{Rh}+\mathrm{ve}$ RBCs from the mother's $\mathrm{Rh}-\mathrm{ve}$ circulation. However, during delivery of the first child, maternal exposure to small amounts of foetal $\mathrm{Rh}+\mathrm{ve}$ blood triggers the mother to prepare anti-Rh antibodies. In subsequent pregnancies with an $\mathrm{Rh}+\mathrm{ve}$ foetus, these maternal Rh antibodies leak across the placenta and destroy foetal RBCs.

Practice Question 4 (Single-Choice MCQ)

Identify the correct sequence of enzymes, zymogens, and ions involved in the blood coagulation cascade following vascular trauma:

(1) Thrombokinase $\xrightarrow{\mathrm{Mg}^{++}}$ Thrombin $\longrightarrow$ Prothrombin $\longrightarrow$ Fibrin

(2) Cascade factors $\xrightarrow{\mathrm{Ca}^{++}}$ Thrombokinase; Prothrombin $\xrightarrow{\text{Thrombokinase}}$ Thrombin; Fibrinogen $\xrightarrow{\text{Thrombin}}$ Fibrin

(3) Cascade factors $\xrightarrow{\mathrm{Na}^{+}}$ Thrombin; Fibrinogen $\xrightarrow{\text{Thrombin}}$ Thrombokinase $\longrightarrow$ Fibrin

(4) Platelets $\longrightarrow$ Fibrin $\xrightarrow{\text{Thrombokinase}}$ Fibrinogen $\xrightarrow{\text{Thrombin}}$ Coagulum

Correct Answer: (2)

Worked Solution: Coagulation is initiated by factors released from injured tissues and stimulated platelets. Through a cascade process involving inactive plasma clotting factors and Ca^{++} ions, the enzyme complex **thrombokinase** is formed. Thrombokinase converts inactive plasma **prothrombin** into active **thrombin**. Thrombin then converts inactive **fibrinogens** in plasma into insoluble **fibrins**, which trap dead and damaged formed elements to form the clot.

Practice Question 5 (Single-Choice MCQ)

How does interstitial fluid (lymph) differ in composition from blood plasma within tissue capillaries?

(1) Lymph has a completely different mineral distribution compared to plasma and contains abundant erythrocytes.

(2) Lymph contains larger plasma proteins and thrombocytes, but lacks water-soluble nutrients and lymphocytes.

(3) Lymph has the same mineral distribution as plasma, but leaves the larger proteins and most of the formed elements behind in the blood vessels.

(4) Lymph cannot transport hormones or absorb fats from intestinal lacteals.

Correct Answer: (3)

Worked Solution: As blood passes through tissue capillaries, water and small water-soluble substances move out into intercellular spaces, leaving the **larger proteins and most of the formed elements** inside the blood vessels. This interstitial fluid (and lymph) has the **same mineral distribution as plasma**, contains specialized lymphocytes, carries nutrients and hormones, and absorbs fats via lacteals in the intestinal villi.

Practice Question 6 (Single-Choice MCQ)

Which of the following pairs of animal groups and their circulatory characteristics is **correctly matched**?

(1) Annelids and Arthropods — Open circulatory system with blood pumped into sinuses

(2) Fishes — 2-chambered heart pumping oxygenated blood directly to body tissues (single circulation)

(3) Amphibians and Lizards — 3-chambered heart with two atria and one ventricle exhibiting incomplete double circulation

(4) Crocodiles — 3-chambered heart with mixing of oxygenated and deoxygenated blood in the single ventricle

Correct Answer: (3)

Worked Solution:

- (1) is incorrect: Arthropods and molluscs have an open circulatory system, whereas **annelids** and chordates have a **closed circulatory system**.
- (2) is incorrect: In fishes, the 2-chambered heart pumps out **deoxygenated blood** to the gills for oxygenation before it goes to body parts.
- (3) is correct: Amphibians and reptiles (except crocodiles) have a 3-chambered heart (2 atria, 1 ventricle) where oxygenated blood from the left atrium and deoxygenated blood from the right atrium mix in the single ventricle (incomplete double circulation).
- (4) is incorrect: **Crocodiles** are an exception among reptiles and possess a **4-chambered heart** like birds and mammals.

Practice Question 7 (Single-Choice MCQ)

Trace the correct anatomical pathway of an action potential from its site of origin to the ventricular myocardium in the human heart:

(1) AVN (right upper corner of RA) $\longrightarrow$ SAN $\longrightarrow$ Bundle of His $\longrightarrow$ Purkinje fibres

(2) SAN (right upper corner of RA) $\longrightarrow$ AVN (lower left corner of RA) $\longrightarrow$ AV bundle (Bundle of His) $\longrightarrow$ Right and Left bundle branches $\longrightarrow$ Purkinje fibres

(3) SAN (left atrium) $\longrightarrow$ AVN (inter-ventricular septum) $\longrightarrow$ Purkinje fibres $\longrightarrow$ Bundle of His

(4) SAN $\longrightarrow$ Purkinje fibres $\longrightarrow$ AVN $\longrightarrow$ AV bundle $\longrightarrow$ Ventricular musculature

Correct Answer: (2)

Worked Solution: The action potential is generated by the **SAN** (pacemaker, located in the right upper corner of the right atrium), travels to the **AVN** (located in the lower left corner of the right atrium close to the atrio-ventricular septum), continues into the **AV bundle (Bundle of His)** passing through the atrio-ventricular septa to emerge on top of the inter-ventricular septum, divides into **right and left bundle branches**, and finally spreads through the ventricular musculature via **Purkinje fibres**.

Practice Question 8 (Single-Choice MCQ)

During the cardiac cycle, what causes the closure of the tricuspid and bicuspid valves, and which heart sound is associated with this event?

(1) Fall in ventricular pressure during ventricular diastole; produces the second heart sound (dub).

(2) Rise in ventricular pressure during ventricular systole causing attempted backflow into atria; produces the first heart sound (lub).

(3) Simultaneous contraction of both atria during atrial systole; produces the first heart sound (lub).

(4) Opening of semilunar valves into the aorta and pulmonary artery; produces the second heart sound (dub).

Correct Answer: (2)

Worked Solution: At the onset of **ventricular systole**, ventricular pressure increases sharply, causing the closure of the **tricuspid and bicuspid valves** due to attempted backflow of blood into the atria. This closure produces the **first heart sound (lub)**. Conversely, the **second heart sound (dub)** is associated with the closure of the **semilunar valves** when ventricular pressure falls during ventricular diastole.

Practice Question 9 (Numerical Application MCQ)

If the duration of a single cardiac cycle in a person is found to be 0.75 seconds and the volume of blood pumped out by the left ventricle per minute (cardiac output) is $5600\,\mathrm{mL}$, what is the person's stroke volume?

(1) $65\,\mathrm{mL}$

(2) $70\,\mathrm{mL}$

(3) $75\,\mathrm{mL}$

(4) $80\,\mathrm{mL}$

Correct Answer: (2)

Worked Solution:

First, calculate the heart rate (number of beats per minute) from the cardiac cycle duration:

$$\text{Heart Rate} = \frac{60 \text{ seconds}}{0.75 \text{ seconds per beat}} = 80 \text{ beats min}^{-1}$$

Next, use the relation $\text{Cardiac Output} = \text{Stroke Volume} \times \text{Heart Rate}$:

$$\text{Stroke Volume} = \frac{\text{Cardiac Output}}{\text{Heart Rate}} = \frac{5600\,\text{mL min}^{-1}}{80\,\text{min}^{-1}} = 70\,\text{mL}$$

Practice Question 10 (Single-Choice MCQ)

In a standard Electrocardiogram (ECG), which of the following statements accurately describes the electrical waves and their physiological correlates?

(1) The P-wave represents repolarisation of the atria leading to atrial diastole.

(2) The QRS complex represents depolarisation of the ventricles, and contraction starts shortly after Q, marking the beginning of systole.

(3) The T-wave represents depolarisation of the ventricles, and the peak of the T-wave marks the beginning of joint diastole.

(4) A standard ECG is recorded by attaching four leads—one to each wrist and one to each ankle.

Correct Answer: (2)

Worked Solution:

- The **P-wave** represents electrical excitation (**depolarisation**) of the atria.
- The **QRS complex** represents **depolarisation of the ventricles**, which initiates ventricular contraction starting shortly after Q and marking the beginning of systole.
- The **T-wave** represents ventricular **repolarisation** (return from excited to normal state), and the **end of the T-wave** marks the end of systole.
- To obtain a standard ECG, a patient is connected with **three electrical leads** (one to each wrist and one to the **left ankle**).

Practice Question 11 (Single-Choice MCQ)

Which of the following statements is **true** regarding human blood vessels and specialized circulatory pathways?

(1) The tunica media (middle layer of smooth muscle and elastic fibres) is comparatively much thicker in veins than in arteries.

(2) The hepatic portal vein carries oxygenated blood directly from the aorta to the intestine.

(3) The pulmonary artery carries deoxygenated blood from the right ventricle to the lungs, whereas pulmonary veins carry oxygenated blood from the lungs to the left atrium.

(4) The coronary system of blood vessels transports blood from the digestive tract to the liver.

Correct Answer: (3)

Worked Solution: In pulmonary circulation, the right ventricle pumps deoxygenated blood into the **pulmonary artery** to the lungs, and oxygenated blood returns via **pulmonary veins** into the left atrium. Note that (1) is false because *tunica media is comparatively thin in veins*; (2) and (4) are false because the **hepatic portal vein** carries blood from the intestine to the liver, while the **coronary system** is exclusively for blood circulation to and from the cardiac musculature.

Practice Question 12 (Single-Choice MCQ)

How do the autonomic nervous system (ANS) and hormones regulate the activity of the myogenic human heart?

(1) Sympathetic neural signals decrease the rate of heartbeat and reduce ventricular contraction strength.

(2) Parasympathetic neural signals decrease the rate of heartbeat, decrease the speed of conduction of action potential, and thereby decrease cardiac output.

(3) Adrenal medullary hormones decrease the cardiac output during stress.

(4) The neural centre moderating cardiac function through the ANS is located in the cerebellum.

Correct Answer: (2)

Worked Solution: A special neural centre in the **medulla oblongata** moderates cardiac function through the ANS. **Sympathetic nerves** and **adrenal medullary hormones** increase heart rate, strength of ventricular contraction, and cardiac output. Conversely, **parasympathetic neural signals** decrease heart rate, decrease the speed of conduction of action potentials, and thereby reduce cardiac output.

Practice Question 13 (Single-Choice MCQ)

Carefully distinguish between the disorders of the circulatory system and select the **correct** statement:

(1) Heart failure is identical to cardiac arrest, which occurs when the heart muscle is suddenly damaged by inadequate blood supply.

(2) Repeated blood pressure checks of $120/80\ \text{mm Hg}$ indicate hypertension and cause damage to brain and kidney tissues.

(3) Coronary Artery Disease (atherosclerosis) is caused by deposits of calcium, fat, cholesterol, and fibrous tissues that narrow the lumen of arteries supplying the heart muscle.

(4) Angina pectoris is characterized by congestion of the lungs and only occurs in elderly males.

Correct Answer: (3)

Worked Solution:

- (3) is correct: **Coronary Artery Disease (CAD) / atherosclerosis** narrows the arterial lumen due to deposits of calcium, fat, cholesterol, and fibrous tissues.
- (1) is incorrect: **Heart failure** (congestive heart failure, marked by lung congestion) means the heart is not pumping blood effectively enough to meet body needs. It is distinct from **cardiac arrest** (when the heart stops beating) and **heart attack** (when heart muscle is suddenly damaged by inadequate blood supply).
- (2) is incorrect: $120/80\ \text{mm Hg}$ is **normal** blood pressure; $140/90\ \text{mm Hg}$ **or higher** indicates hypertension.
- (4) is incorrect: **Angina pectoris** is acute chest pain when insufficient oxygen reaches the heart muscle and can occur in men and women of any age (though more common among the middle-aged and elderly).

7. Common Exam Pitfalls & High-Yield Revision Checklist

Frequent NEET UG Traps to Avoid

- **Plasma vs. Serum:** Never confuse the two—*Serum = Plasma minus clotting factors*.
- **WBC Percentages & Roles:** Students often swap Eosinophils (2–3%, allergic reactions & resist infections) with Basophils (0.5–1%, least abundant, secrete histamine, serotonin, heparin in inflammatory reactions).
- **Universal Donor vs. Recipient:** Group O has *neither A nor B antigens* on RBCs, but has *both anti-A and anti-B antibodies* in plasma (Universal Donor). Group AB has *both A and B antigens* on RBCs, and *nil antibodies* in plasma (Universal Recipient).
- **Crocodile Exception:** Reptiles generally have a 3-chambered heart, but **crocodiles have a 4-chambered heart** with complete double circulation.
- **Valve Locations: Tricuspid** is on the **Right** (between RA and RV); **Bicuspid (Mitral)** is on the **Left** (between LA and LV).
- **Atrial Systole Contribution:** Approximately 70% of ventricular filling happens passively during joint diastole; atrial systole only increases blood flow into the ventricles by about 30%.
- **Heart Failure vs. Heart Attack vs. Cardiac Arrest:**
  - *Heart Failure:* Heart does not pump blood effectively enough to meet body needs (lung congestion).
  - *Heart Attack:* Heart muscle is suddenly damaged by inadequate blood supply.
  - *Cardiac Arrest:* Heart completely stops beating.

8. Concise Final Recap

- **Blood Composition:** 55% Plasma (90–92% water; 6–8% proteins: fibrinogen for clotting, globulins for defense, albumins for osmotic balance) + 45% Formed Elements (RBCs: 5–$5.5\ \text{million mm}^{-3}$; WBCs: 6000–$8000\ \text{mm}^{-3}$; Platelets: 1.5–$3.5\ \text{lakh mm}^{-3}$).
- **Coagulation:** $\text{Thrombokinase (cascade} + \text{Ca}^{++}) \longrightarrow \text{converts Prothrombin to Thrombin} \longrightarrow \text{converts Fibrinogen to Fibrin mesh.}$
- **Cardiac Conduction & Cycle:** Myogenic heart paced by SAN (70–$75\ \text{min}^{-1}$). Cycle duration $= 0.8\ \text{s}$; Stroke volume $= 70\ \text{mL}$; Cardiac output $= 5\ \text{L min}^{-1}$. First sound (*lub*) = closure of tricuspid/bicuspid valves; Second sound (*dub*) = closure of semilunar valves.
- **ECG Waves:** P-wave = atrial depolarisation; QRS complex = ventricular depolarisation; T-wave = ventricular repolarisation.
- **Double Circulation & Regulation:** Pulmonary + Systemic pathways prevent mixing of oxygenated and deoxygenated blood. Moderated by medulla oblongata via ANS (sympathetic increases cardiac output; parasympathetic decreases cardiac output) and adrenal medullary hormones.